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Visualization of a Machine Learning Framework toward Highly Sensitive Qualitative Analysis by SERS

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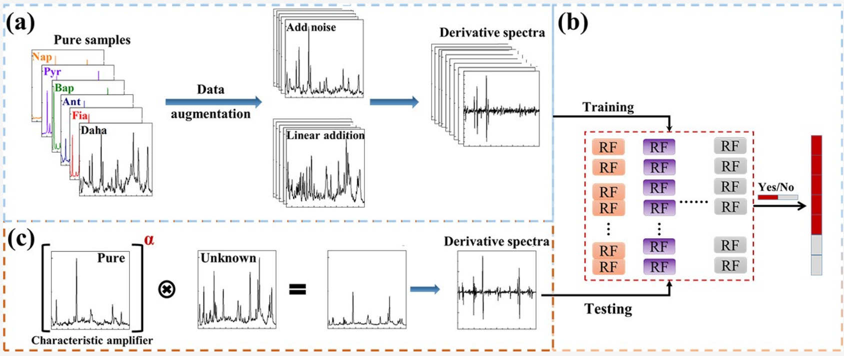

ABSTRACT: Surface-enhanced Raman spectroscopy (SERS), providing near-single-molecule-level fingerprint information, is a powerful tool for the trace analysis of a target in a complicated matrix and is especially facilitated by the development of modern machine learning algorithms. However, both the high demand of mass data and the low interpretability of the mysterious black-box operation significantly limit the well-trained model to real systems in practical applications. Aiming at these two issues, we constructed a novel machine learning algorithm-based framework (Vis-CAD), integrating visual random forest, characteristic amplifier, and data augmentation. The introduction of data augmentation significantly reduced the requirement of mass data, and the visualization of the random forest clearly presented the captured features, by which one was able to determine the reliability of the algorithm. Taking the trace analysis of individual polycyclic aromatic hydrocarbons in a mixture as an example, a trustworthy accuracy no less than 99% was realized under the optimized condition. The visualization of the algorithm framework distinctly demonstrated that the captured feature was well correlated to the characteristic Raman peaks of each individual. Furthermore, the sensitivity toward the trace individual could be improved by least 1 order of magnitude as compared to that with the naked eye. The proposed algorithm distinguished by the lesser demand of mass data and the visualization of the operation process offers a new way for the indestructible application of machine learning algorithms, which would bring push-to-the-limit sensitivity toward the qualitative and quantitative analysis of trace targets, not only in the field of SERS, but also in the much wider spectroscopy world. It is implemented in the Python programming language and is open-source at https://github.com/3331822w/Vis-CAD.

■ INTRODUCTION

The trace detection of multitargets qualitatively and quantitatively is one of the consecutive hotspots in a variety of fields such as biomedicine,[1−4] environmental science,[5] and food safety.[6−8] Surface-enhanced Raman spectroscopy (SERS), originating from Raman spectroscopy and being ignited by localized surface plasmon resonance, has attracted extensive attention owing to the near-single-molecule-level fingerprint information.[9−11] In particular, the qualification and quantitation of SERS, including Raman spectroscopy, have been greatly improved with the rapid development of modern chemometrics.

Up to now, multivariate calibration and machine learning have been the two methods facilitating Raman spectroscopy analysis in a complicated matrix. Multivariate calibration techniques, including the typical multiple linear regression,[12] principal components analysis,[13] partial least-squares regression,[14] ridge regression,[15,16] the recently emerging non-negative least-squares regression,[17,18] non-negative lasso regression, and non-negative elastic net,[19,20] have successfully realized the qualitative and quantitative analysis of mixtures (such as mixture analysis of methanol, ethanol, and acetonitrile).[21] However, due to the small computation of these algorithms, their performance is less than satisfactory in samples where strong molecular interaction was involved, which was partly solved with the introduction of machine learning. For example, early diagnosis,[22,23] bacterial[24]

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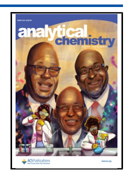

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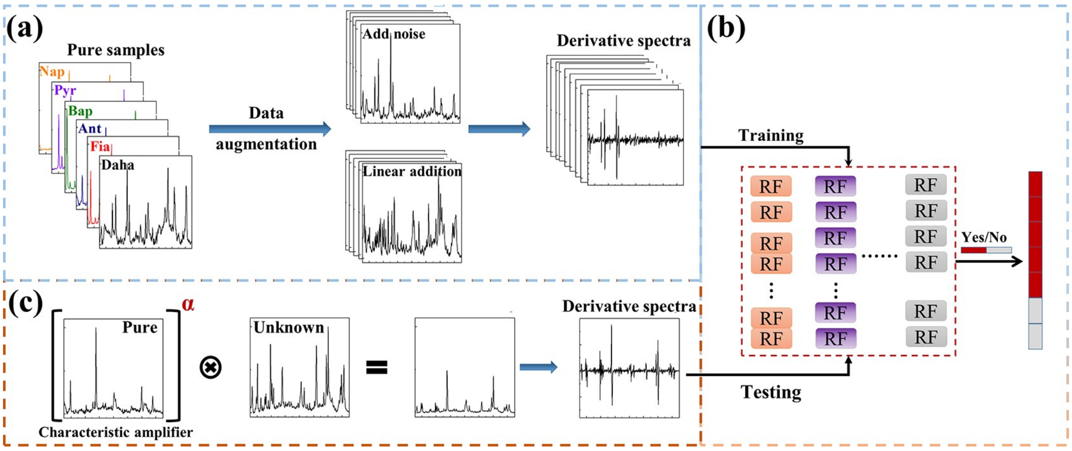

Figure 1. Schematic flowchart of the Vis-CAD: (a) Data preparation and processing. (b) Stochastic forest chain model. (c) Model testing.

and related drug resistance,[25] and quality control of edible oils[26] have been realized by deep neural networks.

Although machine learning algorithms demonstrate a much better performance over those of multivariate calibration ones, there are two open questions: (1) The performance of machine learning remarkably depends on the richness of the provided information, where a sufficient training set is highly demanded, and the related high experimental cost hinders the further improvement of model performance.[27,28] (2) The effectiveness of the trained model depends on the correct features learned. However, the interpretability of the model is normally low accompanied by over fitting, which causes a significantly reduced accuracy of the model toward practical applications compared with the training results.[29,30]

Therefore, aiming to alleviating the experimental burden and visualize the algorithm, we developed a novel algorithm-based framework combining visual random forest, characteristic amplifier, and data augmentation (Vis-CAD), taking the trace analysis of polycyclic aromatic hydrocarbons (PAHs) by SERS as an example. First, data augmentation was applied to create a large number of simulated SERS spectra for a variety of mixtures simply by a small amount of SERS spectra of individual PAHs. The simulated SERS spectra were then used to construct a characteristic amplifier integrated random forest model, which not only successfully realized the qualitative analysis of each PAH in different mixtures with an accuracy of no less than 99% but also notably improved the sensitivity up to at least 1 order of magnitude. More importantly, the interpretability of the model is highly reliable owing to the accurate capture of the characteristic peaks related to the target.

■ EXPERIMENT SECTION

Experiment. In order to make sure that the characteristic Raman signals of each sample are clearly detected, the Raman intensities of the strongest Raman peaks of six PAHs were adjusted at a similar level, where the concentrations were as follows: naphthalene (Nap, 1000 mg/L), pyrene (Pyr, 10 mg/L), benzo[a]pyrene (Bap, 0.452 mg/L), anthracene (Ant, 10.1 mg/L), fluoranthene (Fia, 8.21 mg/L), and dibenzo(a,h)-anthracene (Daha, 50 mg/L). The SERS spectra of different mixtures were obtained by adding these six PAHs step by step (the detailed experimental process can be found in ref 31). The number of SERS spectra, including 6 individual PAHs (20 of each) and 5 mixtures (10 of each), was 170, which could ensure the stability of the result and reduce the interference of external factors such as cosmic rays. The SERS spectra of individuals were used for data augmentation to build the model, and the SERS spectra of mixtures were used to test the reliability of the model.

Methodology. Figure 1 shows the process of the Vis-CAD, which is composed of three steps:

1. *Data preparation and processing.* The set of types of substances is **N**; the flow is shown in Figure 1a.
   - (a) Add low-level (~1% of the strongest signal) Gaussian noise to each individual SERS spectrum to obtain the simulated ones until the required amount (~≥400) is obtained.
   - (b) Randomly select and normalize C_m^n ($m \in [2, \mathbf{N}]$) individual SERS spectra. The selected SERS spectra are multiplied by the random coefficient $[0.1, 1/\mathbf{N}]$ and added to form the simulated SERS spectra of the mixture. Repeat this step until the required amount (~≥1600) is obtained. Similarly, the real mixture spectra can be calculated by multiple linear regression to obtain the proportion of components.
   - (c) According to whether there is a contribution from **I** (**I** is one of six PAHs) to the SERS spectra of the mixture, the spectra obtained from steps a and b are labeled as the negative or positive to the "**I**" individual.
   - (d) In order to reduce the influence of overlapped peaks and background on the model construction, the first derivative of the SERS spectrum in the training set is used for subsequent steps.
2. *Model construction and evaluation of random forest chain.* The flow is shown in Figure 1b.
   - (a) For **I**, the spectra in step 1 are bagged to be divided into in-bag and out-of-bag (OOB). The in-bag spectra are used for training the chain composed of **ten** random forest, and the out-of-bag data is used to evaluate the model performance. The discrimination of **I** is jointly decided by the above **ten** models.
   - (b) For a visualization of the chain model, the process is as follows (implemented through sk-learning in python 3.6): (1) The characteristic importance of random forest is formed by counting the characteristics of nodes used in decision trees. (2) The features used in each random forest are counted to form the vector representing the feature impor-

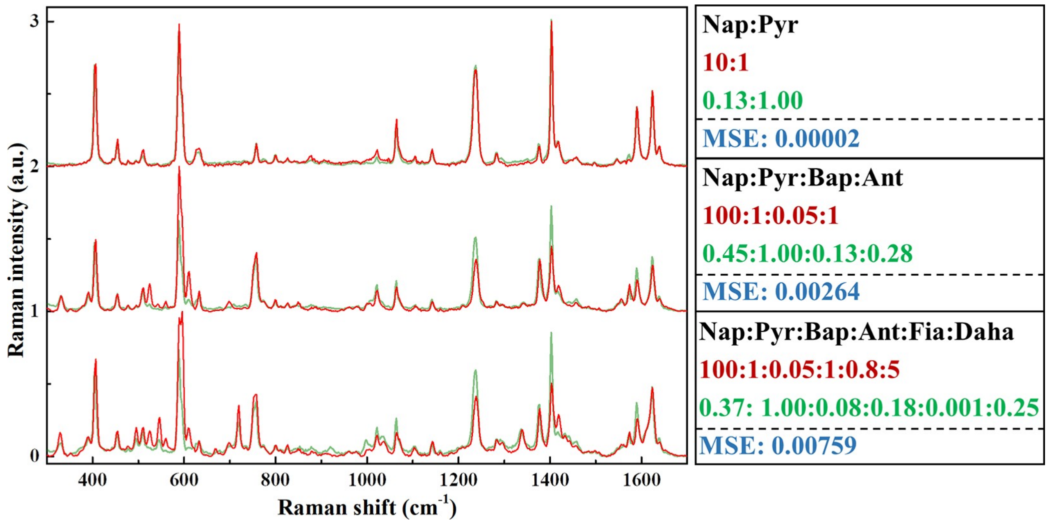

Figure 2. Comparison of SERS spectra of the two-, four-, and six-component simulated mixtures (green line) with that of real mixtures (red line).

tance of the model. (3) Taking the spectrum as the outline, the vector in step 2 is mapped with the color map and used as the filling. Finally, the model visualization is realized.

3. *Identification of the real mixture*. The flow is shown in Figure 1c.
   (a) The characteristic amplifier is generated from the spectrum of **I** (**I** ∈ N). Each point of **I** is amplified by the power of α, and the obtained vector is multiplied by the unknown spectral vector point by point to obtain **I**′. Then, the differential spectrum of **I**′ is put into the chain model for mixture identification.
   (b) The mode of extracting sample features and the causes of misjudgment by the visualization of the model are analyzed.

RESULTS AND DISCUSSION

1. Preparation and Processing. *Reliability of Data Augmentation.* The reliability of data augmentation is demonstrated using the cases of two-, four-, and six-component mixtures as examples. The red lines shown in Figure 2 are the SERS spectra of the mixture, and the green lines are the simulated ones generated by multiple linear regression, using several explanatory variables to roughly predict the outcome of a response variable. The difference between the experimental and simulated ones is evaluated by mean square error (MSE).

Taken for example the two-component mixture, the simplest case, the characteristic Raman peaks of the simulated spectrum are basically the same as those of the experimental spectrum, from the point of the peak position, peak shape, and peak intensity. The slightly different relative peak intensity (MSE = 0.000 02) indicates that the linear sum strategy can effectively reflect the main characteristics of the SERS spectra of the two-component mixture. When it comes to the four-component case, MSE increased rapidly by over 100 times (0.002 64). Both the increasing and decreasing relative Raman intensity could be clearly observed. For example, three peaks located at 524, 611, (C−C and C−H bending vibration from Bap respectively) and 589 cm^{-1} (ring breathing vibration of Pyr) increased, accompanied by the decreased Raman intensity of peaks at 1237 (skeleton torque vibration of Bap and Pyr), 1403 (C−C stretching vibration and skeleton torque vibration of Pyr), and 1589 and 1624 cm^{-1} (C−C stretching vibration of Pyr). With the type of substances in the mixture increased from four to six, the difference between the simulated and the experimental SERS spectrum further increased (MSE = 0.007 59), where a similar tendency was observed as compared to the four-component case with the appearance of Raman peaks from the newly added substance, such as the Raman peak at 719 cm^{-1} (C−C stretching vibration and C−H bending vibration of Daha).

It should be noted that the concentration-dependent SERS spectra of each individual also suffered the changed relative Raman intensity. For example, with the increase of Bap concentration (Figure S1), the relative Raman intensity of SERS peaks at 524, 611, and 1237 cm^{-1} gradually decreased, which may be a result of the strengthened interaction (mainly $\pi-\pi$ stacking effect) between adsorbed molecules with the increased surface coverage brought by the increased concentration in solution. Therefore, we speculate that the difference between the simulated spectrum and the experimental one is attributed to the changed molecular interaction on the surface. Of course, further efforts and endeavors are highly demanded to understand this interfacial mechanism by revealing the adsorption configuration of PAHs and the assignment of Raman peaks.

As mentioned in the Experimental Section, the concentration of PAHs was adjusted to make the Raman intensities of the strongest Raman peaks of six PAHs at a similar level. For example, when [Nap]:[Pyr] = 100:1, the Raman intensity of the strongest Raman peak of Nap at 758 cm^{-1} is pretty close to that of Pyr at 1403 cm^{-1} (Figure S2). However, as shown in Figure 2, for the two-component mixture, the Raman intensity ratio of these two PAHs decreased from 1.03:1 to 0.13:1 (calculated by multiple linear regression). Accordingly, the relative Raman intensities of other characteristic Raman peaks changed. Obviously, the different contribution of Nap and Pyr to the overall SERS spectrum is likely due to the stronger adsorption of the tetracyclic (Pyr) on the Ag NP surface as compared to that of the bicyclic (Nap). It is no wonder that, for the four-component ([Nap]:[Pyr:]:[Bap]:[Ant] = 100:1:0.05:1) and six-component mixtures ([Nap]:[Pyr]:[Bap]:[Ant]:[Fia]:[Daha] = 100:1:0.05:1:0.8:5), the contributions of these PAHs to the SERS spectrum are 0.45:1:0.13:0.28 and

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0.37:1:0.08:0.18:0.001:0.25, respectively, indicating the strengthened impact from the more complicated competitive adsorption on the overall SERS signal. For example, the contribution difference from Pyr and Fia can reach 3 orders of magnitude in the six-component case; the adsorption capacity of these six PAHs on the Ag NP surface is roughly estimated to be in the order Bap > Pyr > Ant > Daha > Nap > Fia according to the concentration-dependent SERS behavior of the mixtures.

In a word, the competitive adsorption and interfacial interaction between adsorbed molecules lead to the observable deviation between the simulated and the experimental SERS spectra of the mixture. However, the low-level MSE confirms the reliability of the data augmentation, which reflects the main information on the measured spectrum.

2. VISUALIZATION AND APPLICATION OF MODEL

2.1. Model Visualization. Table 1 lists a 100% accuracy without false positives or false negatives for the OOB. In order to understand the classification basis of the model avoiding the potential overfitting-induced fake high accuracy, we visualized six PAH identification models as shown in Figure 3. The top and bottom SERS spectra are normalized from those of individual PAH at a certain concentration and the simulated spectra by data augmentation, respectively. The filled Raman peaks represent the features extracted by the model. The darker color, the greater the contribution of the peak used for identifying each PAH in the judge model.

Table 1. Accuracy of Different Models in Simulated and Real Mixtures

	simulated		real		
	unprocessed	derivative	unprocessed	derivative	derivative + CAE
Nap	1	1	0	0.6	1
Pyr	1	1	1	1	1
Bap	1	1	0.2	0.66	1
Ant	1	1	0.2	0.52	1
Fia	1	1	0.6	0.6	0.98
Daha	1	1	0.96	1	1

In general, the model built by the simulated data set can accurately capture the main characteristic Raman peaks of each PAH in the mixed SERS spectra, while the role played by these peaks is significantly different. Taking the identification of Fia as an example, the important degree of the characteristic Raman peaks is in the order of 670 cm^{-1} > 349 cm^{-1} (0.86) ≈ 800 cm^{-1} (0.76) ≈ 1100 cm^{-1} (0.68) > 484 cm^{-1} (0.52) > 560 cm^{-1} (0.46) > 1016 cm^{-1} (0.38) > 1451 cm^{-1} (0.23) > 1422 cm^{-1} (0.18) > 1032 cm^{-1} (0.14), where the number in parentheses represents the important degree being evaluated by the peak area of the SERS peaks in the feature importance vector (Figure S4). The strongest Raman peak of Fia at 670 cm^{-1} does not overlap with Raman peaks of other PAHs and thereby has the highest importance degree in identification. The same is true for the peaks at 349, 800, and 1100 cm^{-1}, while the important degree steadily decreases for the left peaks due to the significant overlapping with the Raman peaks of the other PAHs. For example, due to the overlapping of the 495 cm^{-1} peak of Daha with a similar Raman intensity as the 484 cm^{-1} peak (Fia), the contribution of this peak in the identification of Fia is less than the above peaks. When it comes to the 1032 cm^{-1} peak, its contribution is quite low, attributed to both the low Raman intensity and the serious overlapping with the SERS peaks from Nap, Bap, Ant, and Daha. Furthermore, as extracted from Figure S4, an asymmetric behavior is seen for the SERS peaks with low contribution. For example, the importance degree (0.39) of the left part (low wavenumber range) is less than that (0.66) of the

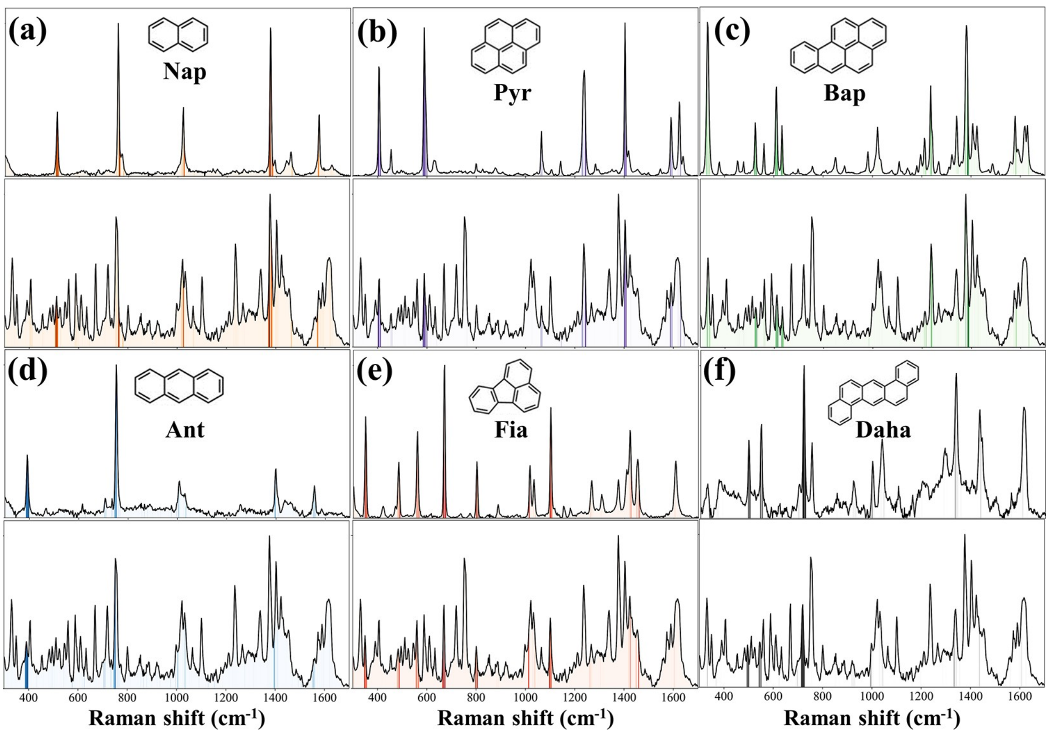

Figure 3. Visualization of six identification models on the basis of the comparison of the SERS spectrum of the simulated mixture with that of 1000 mg/L Nap (a), 10 mg/L Pyr (b), 0.452 mg/L Bap (c), 10.1 mg/L Ant (d), 8.21 mg/L Fia (e), and 50 mg/L Daha (f).

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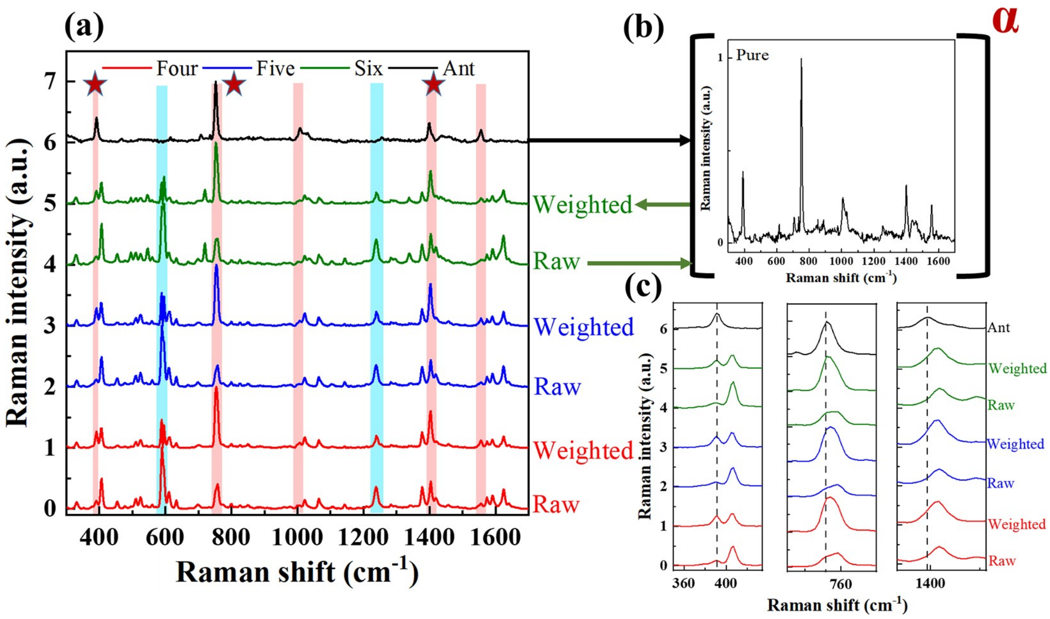

Figure 4. (a) SERS spectra of four-, five-, and six-component mixtures before (raw) and after (weighted) the introduction of a characteristic amplifier. (b) SERS spectrum of Ant as the characteristic amplifier. (c) Zoomed-in section from part a marked by a red star.

right one for the 560 cm^{-1} peak (Fia), which is mainly due to the greater overall impact from the 546 cm^{-1} peak of Daha and the 524 cm^{-1} peak of Bap on the left part of the 560 cm^{-1} peak.

It should be emphasized that a 100% identification accuracy of the model could be obtained for the simulated data set regardless of whether the first derivative is applied or not. However, by visualizing the model, we found that the feature extracted without the first derivative operation could not reflect the distinctive SERS peaks of each PAH (detailed description shown in Figure S5). Furthermore, the increased noise would increase the difficulty of feature extraction (detailed description shown in Figure S6). The inaccurate feature extraction will drastically reduce the accuracy of the model when it comes to the experimentally obtained SERS spectra of the mixture.

2.2. Model Application-Improved Qualification. As shown in Table 1, before the introduction of the first derivative, the identification accuracy in the real sample is very low; in particular, that of Nap is 0. With the application of the first derivative, the identification accuracy is improved significantly, such as those from 0 to 0.6 for Nap, from 0.2 to 0.66 for Bap, from 0.2 to 0.52 for Ant, and from 0.96 to 1 for Daha. Nevertheless, the improved accuracy is still not good enough for these targets, and a further detailed inspection reveals that the misclassifications all belong to the false negative samples.

Taking the identification of Ant as an example, with the increasing types of other PAHs in the measured solution, the identification accuracy of Ant decreases rapidly, from 40% of the four-component cases to 10% of the five-component ones and finally to zero in the six-component mixtures. As has been well discussed in Figure 3d, in the Ant identification model, the importance degree of SERS peaks is in the order of 389 cm^{-1} (1) > 752 cm^{-1} (0.70) > 1398 cm^{-1} (0.30) > 1555 cm^{-1} (0.25) > 1008 cm^{-1} (0.23), and thereby, the first three peaks play the key role in the identification of Ant. However, in a comparison of the SERS spectra of four-component (blue line, raw), five-component (red line, raw), and six-component (green line, raw) mixtures with that of individual Ant (black line) (Figure 5a), the relative Raman intensity of the separate peak at 389 cm^{-1} in the SERS spectrum of the four-component mixture is very weak, not to mention that the next two peaks located at 752 and 1398 cm^{-1} overlap seriously with the SERS peaks from Nap (757 cm^{-1}), and Pyr (1403 cm^{-1}) or Bap (1407 cm^{-1}), respectively. The overwhelmed SERS signal of Ant in the mixed SERS spectra is most possibly a result of the competitive adsorption from other PAHs, as well-discussed in Figure 1. The greater the number of types of PAHs in a mixture, the stronger the competitive adsorption and the lesser the contribution of Ant to the mixed SERS spectrum. To overcome the negative effect from the competitive adsorption, we designed a characteristic amplifier (CAE) based on the SERS spectrum of the individual.

The weighted lines in Figure 4 are the raw and processed SERS spectra of four-component (red line), five-component (blue line), and six-component (green line) mixtures. It is determined that not only the separated SERS peak of Ant at 389 cm^{-1} but also the overlapped SERS peaks at 752 or 1398 cm^{-1} showed a selectively upward trend in the relative Raman intensity. Meanwhile, the relative Raman intensities of SERS peaks from other PAHs, such as the peaks at 589 and 1238 cm^{-1} (marked in blue), showed a downward trend. Therefore, the identification accuracy of Ant was improved significantly from 52% to 100%, free of false negatives and false positives as well.

As shown in Table 1, by selecting appropriate parameters of CAE, the 100% qualitative identification of the left four individual PAHs was also realized in the mixtures, except 98% for the Fia case, where one false positive is for a four-component mixture free of Fia. By revisiting the SERS spectrum of this sample, we determined that the two characteristic SERS peaks located at 349 and 672 cm^{-1}, having high importance degree for Fia identification, are very weak but discernible here (Figure S9). Therefore, the CAE operation further amplified the SERS intensity of these two peaks and resulted in a false positive, in comparison with the SERS signal of Fia, which is always the weakest in any mixture case (Figure 2) due to the strong competitive adsorption from other PAHs.

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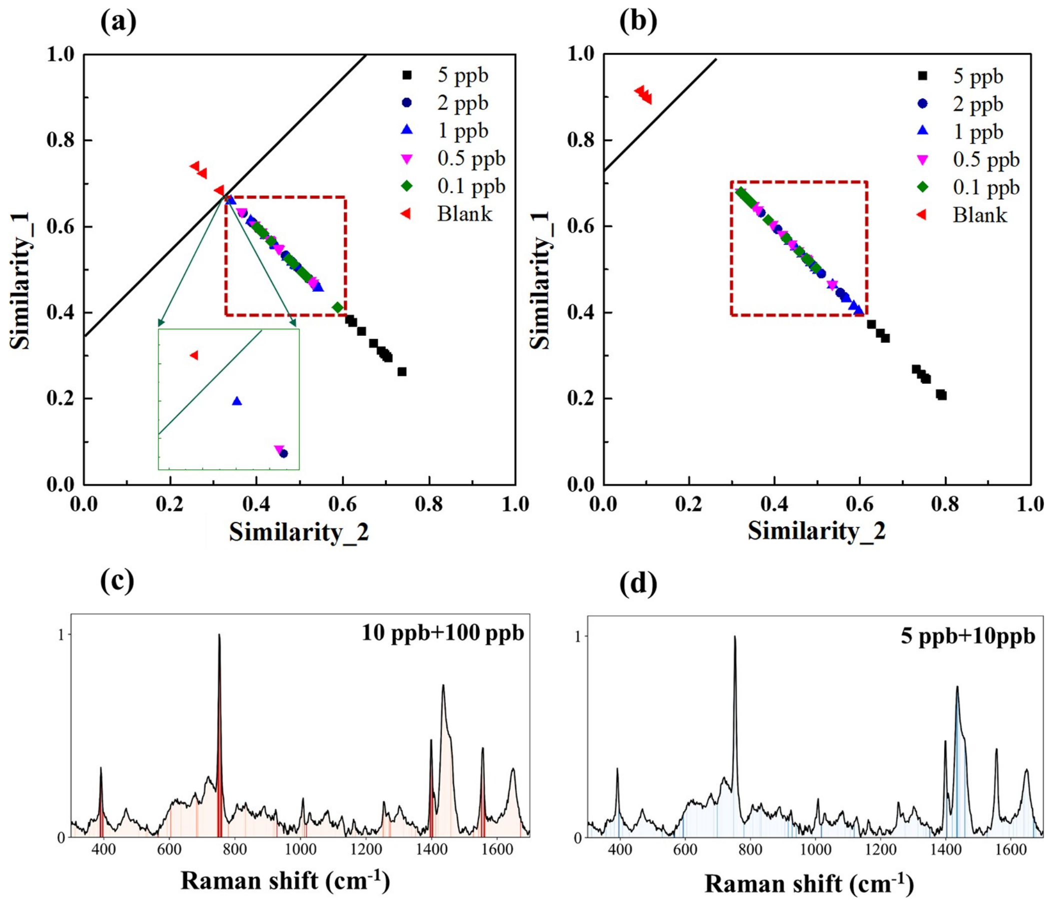

Figure 5. Ant concentration-dependent similarity extracted from the high-concentration model (a) or low concentration model (b). The feature used for each model is visualized in panels c and d, respectively.

It should be noted that the amplified factor α can be adjusted according to demand of classification, which is optimized by a grid search with the criteria of obtaining the highest accuracy free of false positives. The details are discussed in Figure S7 with Fia as an example.

2.3. Model Application-Improved Sensitivity. The high identification of individual PAHs in mixture further offered a chance for improving SERS sensitivity. Here, the detection of Ant was used as an example with the lowest detectable concentration of ~5 ppb,[31] where the SERS signal of Ant is extremely weak (Figure S10).

By defining the SERS spectra of Ant at the two higher concentrations (10 and 100 ppb) or the two lower concentrations (5 and 10 ppb) as the positive set and the SERS spectra of the blank being added with random noise as the negative set, respectively, the two visual identification models of Ant were established. The outputted Ant concentration-dependent similarity vectors are shown in Figure 5, where similarity 1 and similarity 2 refer to the similarity with negative and positive samples, respectively. Clearly, the distinctive difference between the similarity vector of the SERS spectra of the blank and that of the 5 ppb Ant implies that both models achieve 100% identification of 5 ppb (positive) and the blank (negative).

When it comes to the lower concentrations of Ant from 2 ppb down to 0.1 ppb, the similarity vectors were nonuniformly distributed in between as an intermediate state (marked in a dashed frame), implying the hard-to-tell difference among the SERS spectra under these concentrations. Meanwhile, different to the almost jointed boundary of the dashed frame to the blank one for the higher-concentration model (Model 1 in Figure 5a), the frame is far away from the blank one for the lower concentration model (Model 2 in Figure 5b), indicating the distinguishable difference of SERS spectra between the blank and the one containing Ant even down to 0.1 ppb. The significantly improved sensitivity from 5 ppb (by the naked eye) to 0.1 ppb inspired us to revisit the characteristic Raman signal extracted by these two models, and the visualized results are shown in Figure 5c,d.

Different to Model 1 where the characteristic SERS peaks of Ant (Figure 5c) were used for determining similarity, the SERS peaks of citrate[31] (Figure 5d), the surface protector of the Ag NPs, were the dominant ones for Model 2. The higher discrimination by capturing the Ant concentration-dependent SERS signal of citrate in Model 2 over that by Ant itself in Model 1 offers a more distinctively high sensitivity. However, everything has two sides. The indirect detection provided higher sensitivity but lost selectivity. Therefore, when facing trace detection in a complex matrix, Model 1 is preferred along with the necessary sample pretreatment. Furthermore, the influence of potential Raman shift drifting on the Vis-CAD was also tentatively explored. By adding 3 or 6 cm^{-1} Raman shifts to the SERS spectra of 5 ppb Ant, the accuracy of Model 1 drastically decreased from 1 to 0.83 and 0.47, respectively. The decreased accuracy is mainly attributed to the highly sensitive CAE tolerance on peak position, although the identification by multiple sampling points in a peak (Figure 5c) endows the tolerance of Raman shift drifting. Therefore, before the

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identification of trace targets in a matrix, the potential peak position drifting should either be avoided by instrument calibration in advance or be involved in the process of data augmentation with the amplifier as well.

In principle, the advantage of accurately capturing the target characteristics of the random forest algorithm disables itself the power of an overall view of the whole SERS spectra. Therefore, one cannot use the SERS characteristic Raman peaks of both Ant and citrate simultaneously for the determination of the push-to-the-limit detectable concentration. Alternatively, a deep learning algorithm, such as a convolutional neural network (CNN), endowed with fine-grained learning through the data, may capture all features, and the related work is undergoing and will be discussed elsewhere.

■ CONCLUSION

With the combination of visual random forest, characteristic amplifier, and data augmentation, a machine-learning-based algorithm framework, simplified as Vis-CAD, toward a highly selective and sensitive SERS analysis with trustworthy interpretability was realized. (1) For *data augmentation*, creating a mass database of SERS spectra of a variety of mixtures by limited SERS spectra of individuals was demonstrated successfully with a high similarity between the simulated and real spectra. (2) The *visualization* of the importance degree of characteristic SERS peaks used for identification clearly displayed the reliability of the random forest. (3) The introduction of a *characteristic amplifier* based on the SERS spectra of individuals precisely and effectively amplified the tiny SERS signal of the determined individual overwhelmed in the SERS spectrum of mixture, due to the unavoidable competitive adsorption. Therefore, not only was the qualitative identification of an individual PAH in a mixture realized with an accuracy of no less than 99%, but also, a sensitivity improved by more than 1 order of magnitude was achieved. More importantly, with the replacement of the random forest algorithm by the more precise deep-learning algorithms, the push-to-the-limit qualitative and quantitative SERS analysis is highly promising.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.2c01450.

PAHs spectra for the model; details of model visualization vector; description of the role of the first derivative in the model; influence of amplifier parameters; effect of the amplifier on overall data distribution (PDF)

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All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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